



Physiological Validation of Affective Arousal for Psycholinguistic Norms

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Introduction

Emotionality in Psycholinguistics

- Valence, arousal, and dominance are common measures of emotionality in psycholinguistic research (Warriner et al., 2013).
- Some sources explicitly separate subjective affective arousal from physiological arousal (APA, 2018; Sander, 2025; Satpute et al., 2019).
- Others assume they are related (Basu et al., 2016; Soares et al., 2012; Zhang et al., 2025).

Research Gap

- No word-level validation of arousal current exists
- Pupil diameter, heart rate and skin conductance have been shown to correlate with affective arousal for affective images (Bradley et al., 2008).
- Only pupil diameter is a measure of autonomic arousal fast enough to use at the individual word level.

Key Research Question

Do psycholinguistic arousal ratings reflect physiological arousal during semantic processing?

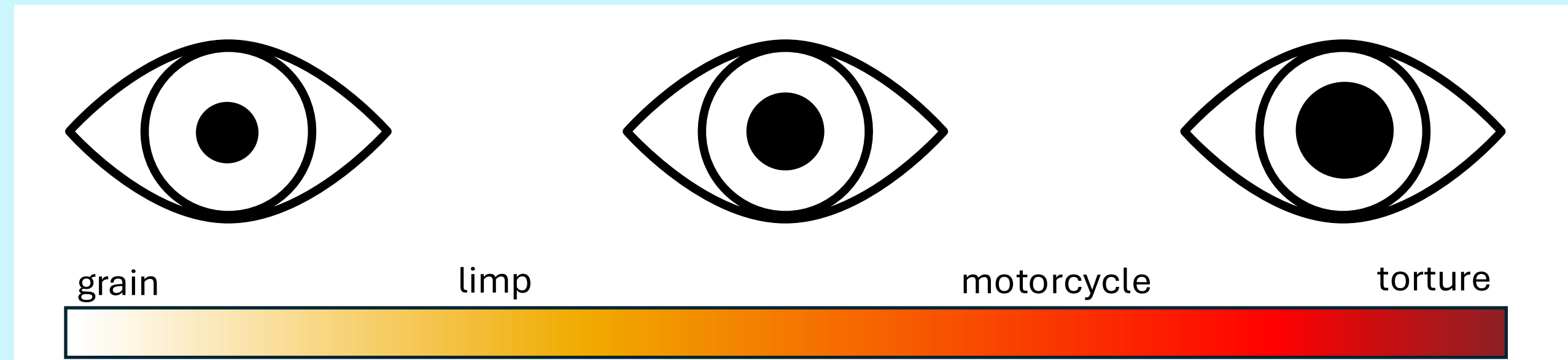


Figure 1. H1: pupil diameter will correlate with subjective psycholinguistic ratings

Hypotheses

- H₁** - Words with higher affective arousal ratings will produce larger pupil dilations during lexical processing tasks.
- H₂** - The relationship between subjective and physiological arousal will strengthen with deeper semantic processing.
- H₃** - Physiological arousal will facilitate RT above a baseline model of lexical predictors (length, OLD20, prevalence).

Outcome	Interpretation
H ₁ supported	Arousal norms reflect embodied physiological experience
H ₁ rejected	Arousal ratings capture another emotional construct
H ₂ supported	Physiological arousal increases with semantic processing
H ₂ rejected	Physiological arousal not impacted by semantic processing
H ₃ supported	Physiological arousal impacts language processing
H ₃ rejected	Physiological arousal does not impact language processing

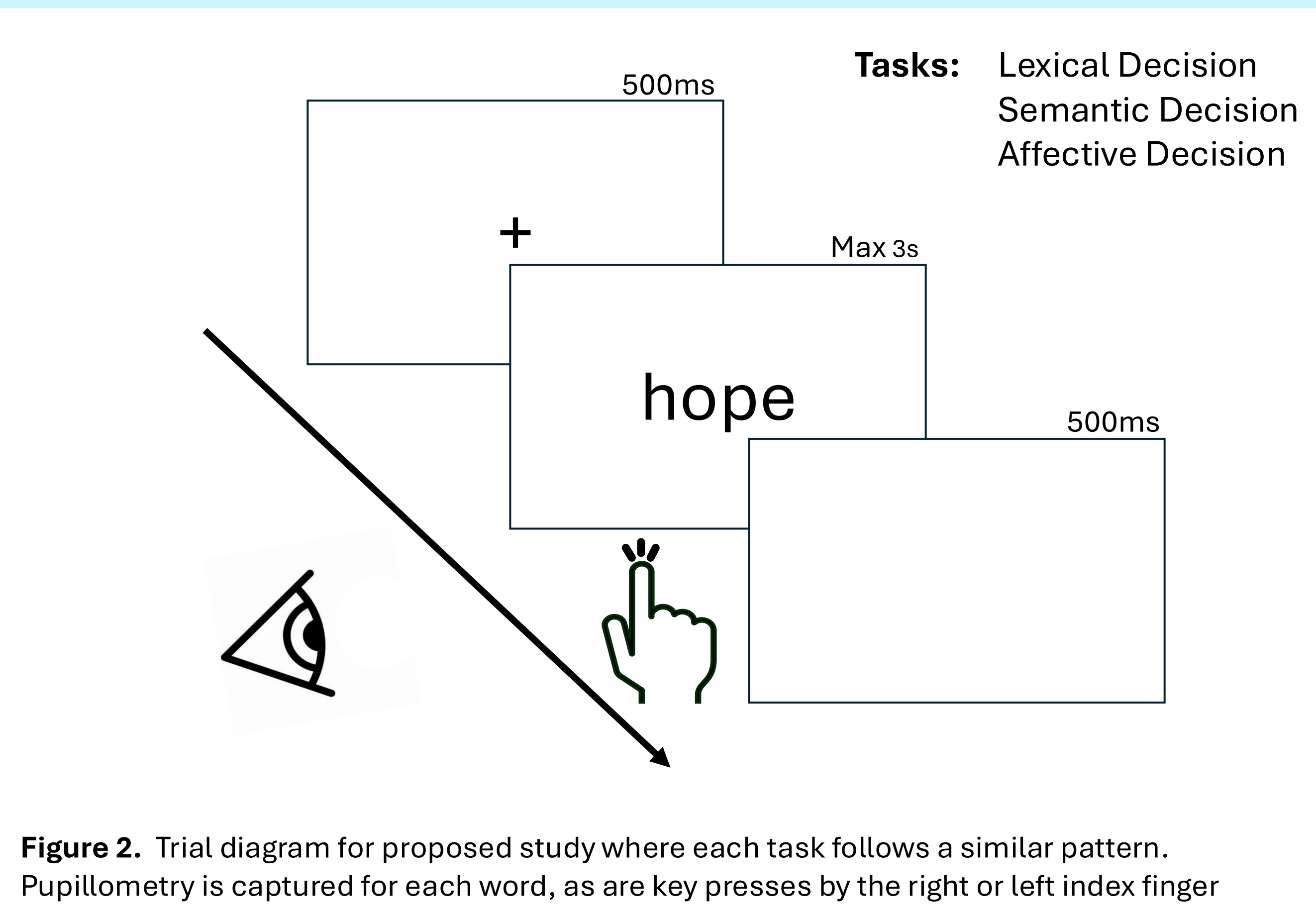


Figure 2. Trial diagram for proposed study where each task follows a similar pattern. Pupillometry is captured for each word, as are key presses by the right or left index finger

Methods

Stimuli

Words from Warriner's norms split across 3 word lists of 65 words each plus 5 calibration words. Each list is balanced for affective and psycholinguistic norms. Low prevalence words are excluded.

Participants

Convenience sampling will be used to recruit nmin 54 to nmax 108 participants. Final n determined by sequential Bayesian hypothesis testing until $BF_{10} \geq 5$ or $\leq 1/5$ for H_1 .

Tasks

Task	Judgement	Semantic Processing
Lexical Decision	Word – Not word	Low
Semantic Decision	Concrete – Abstract	High
Affective Decision	Emotive – Not emotive	High / Affective

- All participants complete all tasks and see all word lists
- Each task consists of trials like those seen in Figure 2 where the participant is presented repeatedly with a words and asked to make the appropriate judgement.
- Pupillometry is captured from the onset of each word until a decision is made.
- Making these judgements participants access their mental representation of the word meaning, allowing us to associate physiological measures with each word.

Plan for Analysis

Item-Level Validation

- Removing sources of participant variance, measures of physiological arousal will be normalized per participant before calculating item-level means per word and task
- Linear regression will be run with subjective arousal norms as dependent variable vs. mean physiological arousal and dummy coded task
 - Inclusion of item-level means gives evidence for H_1 .
 - Inclusion of dummy coded tasks gives evidence for H_2 .

Trial-Level effect on RT

- A Bayesian generalized linear mixed effect model will be built to predict RT using only baseline psycholinguistic variables (dependent variable: RT per trial. fixed effects: psycholinguistic controls, arousal. random effects: participants and items). Two models will then be based on this;
 - a) Inclusion of pupil measurements gives evidence for the effect of Physiological arousal on RT (H_3)
 - b) Inclusion of of subjective ratings taken from existing norms gives evidence for the effect of rated arousal on RT
- The models will then be compared to determine whether there is any difference that may suggest the two measures of arousal capture different underlying constructs

Expected Results

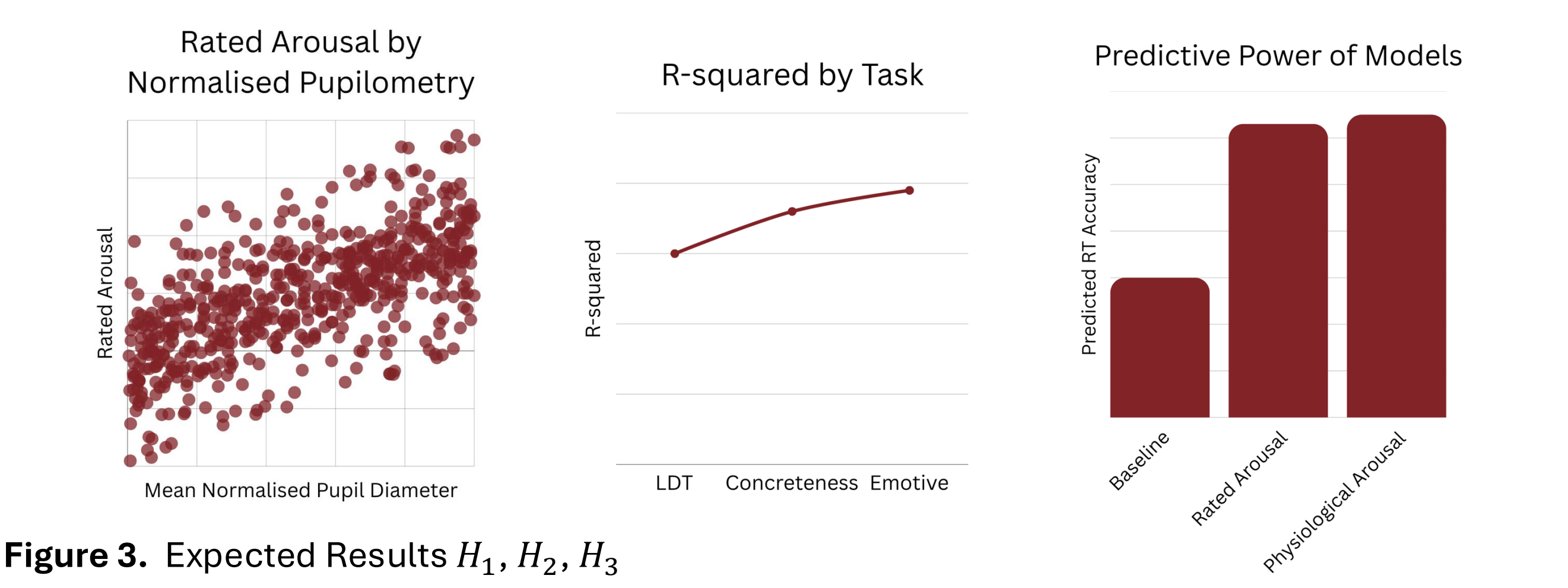


Figure 3. Expected Results H_1 , H_2 , H_3

Conclusion

This study will help clarify whether psycholinguistic arousal ratings capture physiological arousal as measured by pupil diameter, validating a common affective psycholinguistic measure

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